

WASP-36b

R-1552

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_220121 · created 2026-08-01T20:06:39+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459600.8151 +/- 0.0045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.035 +/- 0.048
Transit depth (Rp/Rs)^2	0.12 +/- 0.34 [%]
Orbital Inclination (inc)	81.82 +/- 2.94
Airmass coefficient 1 (a1)	3.35 +/- 1.68
Airmass coefficient 2 (a2)	-0.58 +/- 1.65
Scatter in the residuals of the lightcurve fit is	52.11 %
Best Comparison Star	#3 - [461, 95]
Optimal Aperture	0.87
Optimal Annulus	6.75
Transit Duration (day)	0.04 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.035 ± 0.048 vs 0.1368 (deviation 2.04σ)
Duration fit vs published	0.04 d vs 0.0758 d (deviation 1.38σ)
Mid-time O–C	6.6 min
Depth SNR	0.7
Residual scatter	52.11 %

Light curve

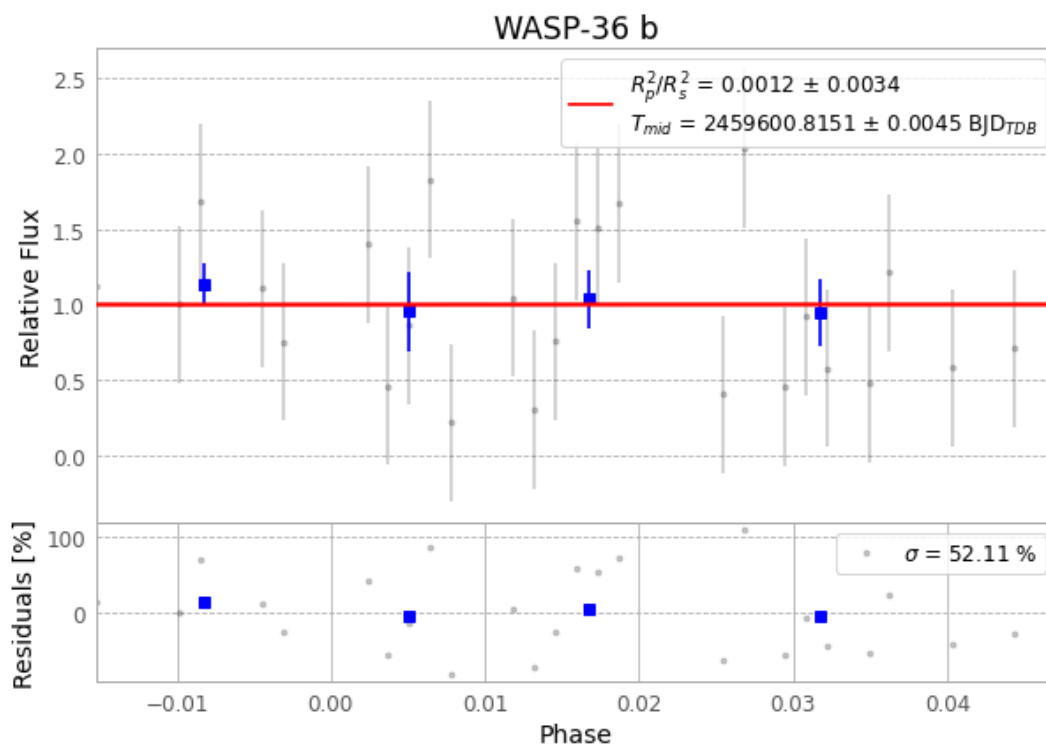


Figure 1 — FinalLightCurve_WASP-36 b_2022-01-20.png (SHA-256 66038e042194bf94...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [412, 439], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2cfba584d98f04a9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

57 FITS frames (37 MB), WASP-36220121062412.FITS → WASP-36220121091512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-220121.log (4123cb75663b...), FinalLightCurve_WASP-36 b_2022-01-20.png (66038e042194...), FinalLightCurve_WASP-36 b_2022-01-20.csv (157582331c87...)

Compute resources

Wall time 135.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 20:06Z.